THE CLASSICAL IMMERSED BOUNDARY METHOD

Mathematical formulation and numerical considerations

Laura A. Miller

Department of Mathematics, University of Arizona

IBFE Tutorials • Module 1

Eulerian fluid

$$\boldsymbol{u}(\boldsymbol{x},t),\ p(\boldsymbol{x},t),\ \boldsymbol{f}(\boldsymbol{x},t)$$

Lagrangian structure

$$\boldsymbol{X}(q,t),\ \boldsymbol{U}(q,t),\ \boldsymbol{F}(q,t)$$

AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

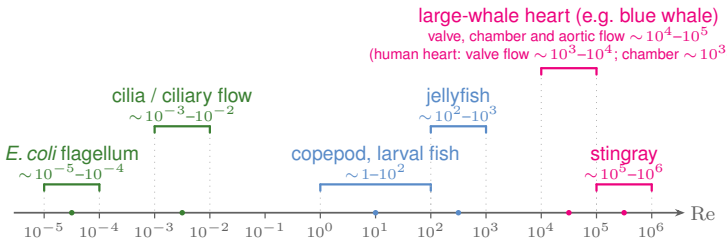
FLUID–STRUCTURE INTERACTION SPANS MANY SCALES

Higher Reynolds number

- whale heart: chambered pumping
- stingray: flapping locomotion

Lower Reynolds number

- respiratory cilia: microscopic pumping
- *E. coli*: flagellar locomotion



Schematic: orders of magnitude, instructor's course notes.

$\mathrm{Re} = UL/\nu$ with L the cell, appendage, valve, or body length and U a typical speed; brackets show routine motion. Copepod escape jumps and late larval fish reach $\sim 10^3$; jellyfish span ~ 10 (ephyrae) to $\sim 10^4$ (large medusae). Compare the table in S. Vogel, *Life in Moving Fluids*, 2nd ed. (Princeton, 1994).

The same conservation laws apply across these examples; the Reynolds number identifies the balance between inertia and viscosity.

MODELING THE FLUID: INCOMPRESSIBLE NAVIER–STOKES

Let $\boldsymbol{u}(\boldsymbol{x}, t)$ be velocity and $p(\boldsymbol{x}, t)$ pressure. For constant density ρ and dynamic viscosity μ,

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = -\nabla p + \mu\Delta\boldsymbol{u} + \boldsymbol{f}, \qquad \nabla\cdot\boldsymbol{u} = 0.$$

Momentum balance

- local and advective acceleration
- pressure and viscous forces
- immersed-structure force $\boldsymbol{f}$

Incompressibility

- velocity is divergence free
- pressure enforces the constraint
- density is transported as a constant

NONDIMENSIONAL NAVIER–STOKES EQUATIONS

Choose characteristic length L and speed U:

$$\boldsymbol{x} = L\boldsymbol{x}^*, \qquad t = \frac{L}{U}t^*, \qquad \boldsymbol{u} = U\boldsymbol{u}^*, \qquad p = \rho U^2 p^*.$$

With $\boldsymbol{f} = (\rho U^2/L)\boldsymbol{f}^*$ and $\nabla = L^{-1}\nabla^*$, every term of the dimensional momentum equation carries a factor $\rho U^2/L$ except the viscous term, which carries $\mu U/L^2$. Dividing by $\rho U^2/L$:

$$\frac{\partial \boldsymbol{u}^*}{\partial t^*} + (\boldsymbol{u}^*\cdot\nabla^*)\boldsymbol{u}^* = -\nabla^* p^* + \frac{1}{\mathrm{Re}}\Delta^*\boldsymbol{u}^* + \boldsymbol{f}^*, \qquad \nabla^*\cdot\boldsymbol{u}^* = 0,$$

$$\boxed{\mathrm{Re} = \frac{\rho U L}{\mu} = \frac{UL}{\nu}}, \qquad \nu = \frac{\mu}{\rho}.$$

The ratio of the viscous to the inertial coefficient is $(\mu U/L^2)/(\rho U^2/L) = \mu/(\rho U L) = 1/\mathrm{Re}$. The pressure scale ρU^2 is the inertial choice; for $\mathrm{Re} \ll 1$ the viscous scale $p = (\mu U/L)p^*$ is the natural one.

REYNOLDS NUMBER AND DYNAMIC SIMILARITY

$$\mathrm{Re} = \frac{\text{inertial scale}}{\text{viscous scale}} = \frac{\rho U L}{\mu}.$$

- $\mathrm{Re} \ll 1$: viscosity dominates.
- $\mathrm{Re} \gg 1$: inertia is important.
- Changing size, speed, density, or viscosity changes the regime.

Similarity needs more than one number.

Matching Re supports dynamic similarity only when the nondimensional geometry, boundary motion, constitutive laws, and other relevant parameters also match.

DESCRIBING THE IMMERSED BOUNDARY

Classical interfacial model

- A material coordinate $q \in \mathcal{U} \subset \mathbb{R}$ labels a massless curve.
- $\Gamma_t = \boldsymbol{X}(\mathcal{U}, t)$ is its current position.
- Elastic energy may resist stretching or bending.
- Constitutive parameters may vary in space or time.

Rigid or prescribed motion

- Stiff elasticity can approximate rigidity, but creates a stiff numerical problem.
- Target-point penalties impose motion approximately.
- Exact constraints require unknown reaction forces.

> Immersed boundaries are represented through forces; they need not describe a passive elastic material.

EULERIAN FLUID, LAGRANGIAN STRUCTURE

Eulerian description

The fixed Cartesian grid carries

$$\boldsymbol{u}(\boldsymbol{x},t), \qquad p(\boldsymbol{x},t), \qquad \boldsymbol{f}(\boldsymbol{x},t).$$

The fluid equations are solved at spatial locations $\boldsymbol{x} \in \Omega$.

Lagrangian description

The moving curvilinear mesh carries

$$\boldsymbol{X}(q,t), \qquad \boldsymbol{U}(q,t), \qquad \boldsymbol{F}(q,t).$$

The material label q follows the structure.

$\boldsymbol{F} \overset{\text{spread}}{\longrightarrow} \boldsymbol{f}$, $\boldsymbol{u} \overset{\text{interpolate}}{\longrightarrow} \boldsymbol{U} = \partial_t \boldsymbol{X}$. The grids do not conform to one another.

THE GOVERNING COUPLING EQUATIONS

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = -\nabla p + \mu\Delta\boldsymbol{u} + \boldsymbol{f}, \qquad \nabla\cdot\boldsymbol{u} = 0,$$

$$\boldsymbol{f}(\boldsymbol{x},t) = \int_{\mathcal{U}} \boldsymbol{F}(q,t)\delta^{(2)}(\boldsymbol{x} - \boldsymbol{X}(q,t))\, dq,$$

$$\frac{\partial \boldsymbol{X}}{\partial t}(q,t) = \boldsymbol{U}(q,t) = \int_{\Omega} \boldsymbol{u}(\boldsymbol{x},t)\delta^{(2)}(\boldsymbol{x} - \boldsymbol{X}(q,t))\, d\boldsymbol{x},$$

$$\boldsymbol{F}(q,t) = -\frac{\delta E[\boldsymbol{X},t]}{\delta \boldsymbol{X}}(q,t).$$

These equations require compatible initial data and fluid boundary conditions. The scope here is a two-dimensional, massless elastic curve with no slip.

Sources: Peskin (2002); Griffith and Patankar (2020).

ELASTIC ENERGY DETERMINES THE FORCE

The variational definition fixes the sign:

$$\delta E[\boldsymbol{X}, t; \boldsymbol{\eta}] = -\int_{\mathcal{U}} \boldsymbol{F}(q,t) \cdot \boldsymbol{\eta}(q)\, dq, \qquad \boxed{\boldsymbol{F} = -\frac{\delta E}{\delta \boldsymbol{X}}}.$$

Typical contributions

- stretching: depends on $|\partial_q \boldsymbol{X}|$
- bending: depends on curvature
- active terms: prescribe a preferred configuration or tension

> $\boldsymbol{F}(q,t)\, dq$ is the force that material segment dq applies to the fluid. Explicit time dependence permits prescribed changes in the constitutive model.

TARGET POINTS IMPOSE MOTION APPROXIMATELY

Let $\boldsymbol{Y}(q,t)$ be the desired trajectory while $\boldsymbol{X}(q,t)$ still moves with the fluid:

$$E_{\text{target}}[\boldsymbol{X},t] = \frac{\kappa}{2}\int_{\mathcal{U}} |\boldsymbol{X}(q,t) - \boldsymbol{Y}(q,t)|^2 \, dq, \qquad \boldsymbol{F}_{\text{target}} = \kappa(\boldsymbol{Y} - \boldsymbol{X}).$$

- Finite κ allows displacement error.
- Damping may also penalize the velocity mismatch.
- Increasing κ can reduce tracking error only while the discretization remains stable and resolved.
- An exact prescribed-motion constraint instead solves for a Lagrange-multiplier force.

A REPRESENTATIVE EXPLICIT TIME STEP

Given $\boldsymbol{X}^n$ and $\boldsymbol{u}^n$:

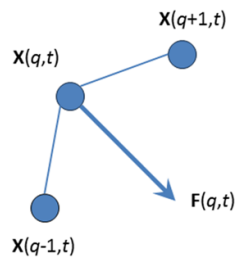
1. compute $\boldsymbol{F}^n = -\delta E[\boldsymbol{X}^n, t^n]/\delta \boldsymbol{X}$;
2. spread $\boldsymbol{f}^n = S[\boldsymbol{X}^n]\boldsymbol{F}^n$;
3. advance incompressible Navier–Stokes to obtain $\boldsymbol{u}^{n+1}$ and p^{n+1};
4. interpolate $\boldsymbol{U}^{n+1} = J[\boldsymbol{X}^n]\boldsymbol{u}^{n+1}$;
5. update $\boldsymbol{X}^{n+1} = \boldsymbol{X}^n + \Delta t\, \boldsymbol{U}^{n+1}$.

> Predictor–corrector, midpoint, and implicit schemes evaluate terms at different time levels. The ordering and every configuration used by S and J must be stated explicitly.

FORCE SPREADING: CONTINUUM PICTURE

$$\boldsymbol{f}(\boldsymbol{x},t) = \int_{\mathcal{U}} \boldsymbol{F}(q,t)\delta^{(2)}(\boldsymbol{x} - \boldsymbol{X}(q,t))\; dq$$

- The singular Eulerian force is supported on Γ_t.
- $\boldsymbol{F}$ is a density with respect to the chosen material coordinate q.
- Integrating $\boldsymbol{f}$ over Ω recovers the total structural force.



Force carried by a Lagrangian point

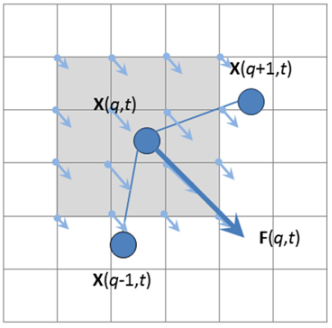
Schematic: instructor's course notes.

FORCE SPREADING: DISCRETE PICTURE

At Eulerian locations $\boldsymbol{x_i}$, use Lagrangian quadrature points q_m with weights w_m:

$$\boldsymbol{f_i}^n = \sum_m \boldsymbol{F}_m^n \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m^n)\, w_m .$$

- $w_m = \Delta q$ only for equally spaced nodal quadrature.
- The support of δ_h determines which grid samples receive force.
- On a staggered grid, each force component uses its own face-centered locations.



A regularized kernel distributes the force locally

Schematic: instructor's course notes.

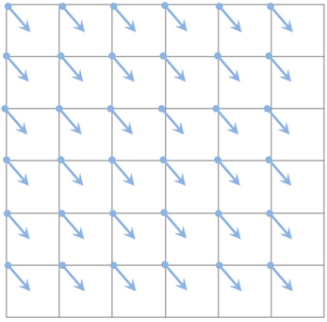
VELOCITY INTERPOLATION: CONTINUUM PICTURE

$$\boxed{\boldsymbol{U}(q,t) = \int_{\Omega} \boldsymbol{u}(\boldsymbol{x},t)\delta^{(2)}(\boldsymbol{x} - \boldsymbol{X}(q,t))\ d\boldsymbol{x}}.$$

For a continuous velocity field,

$$\partial_t \boldsymbol{X}(q,t) = \boldsymbol{U}(q,t) = \boldsymbol{u}(\boldsymbol{X}(q,t),t).$$

> This is the immersed-boundary form of no slip: every material point moves at the local fluid velocity.



Velocity data on the Cartesian grid

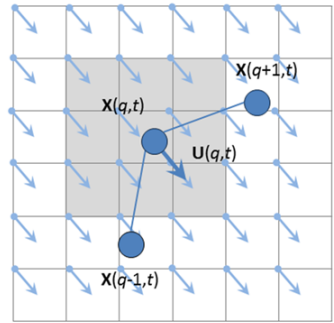
Schematic: instructor's course notes.

VELOCITY INTERPOLATION: DISCRETE PICTURE

For uniform two-dimensional spacing h,

$$\boxed{\boldsymbol{U}_m^n = \sum_{\boldsymbol{i}} \boldsymbol{u}_{\boldsymbol{i}}^n \delta_h(\boldsymbol{x}_{\boldsymbol{i}} - \boldsymbol{X}_m^n)\, h^2}.$$

- The kernel computes a local weighted average.
- Use the same configuration and kernel consistently in both transfers.
- Interpolated marker velocities update the structure.



Eulerian velocity sampled at IB points

Schematic: instructor's course notes.

QUADRATURE WEIGHTS CARRY THE MATERIAL MEASURE

Eulerian sum

$$\sum_{i}(\cdot)\, h^2$$

approximates an integral over physical space.

Lagrangian sum

$$\sum_{m}(\cdot)\, w_m$$

approximates an integral over material coordinate q.

Why the distinction matters

The weights w_m need not be equal and do not have to equal h. In IBFE, interaction quadrature can be refined independently of finite-element nodes and of the Eulerian grid.

Dropping w_m changes the magnitude and units of the transferred force.

SPREADING AND INTERPOLATION FORM AN ADJOINT PAIR

Write $\boldsymbol{f} = S[\boldsymbol{X}]\boldsymbol{F}$ and $\boldsymbol{U} = J[\boldsymbol{X}]\boldsymbol{u}$. With the same kernel and consistent weights,

$$\boxed{\sum_{\boldsymbol{i}} \boldsymbol{f_i} \cdot \boldsymbol{u_i}\, h^2 = \sum_{m} \boldsymbol{F}_m \cdot \boldsymbol{U}_m\, w_m} \qquad \Longleftrightarrow \qquad J[\boldsymbol{X}] = S[\boldsymbol{X}]^*.$$

Physical meaning

- Both grids compute the same instantaneous power transfer.
- The interaction operators add no artificial power source.

> This identity does not by itself guarantee stability of the complete spatial and temporal discretization.

REGULARIZED DELTA KERNELS

On a uniform two-dimensional grid,

$$\boxed{\delta_h(\boldsymbol{x}) = \frac{1}{h^2}\phi\Big(\frac{x_1}{h}\Big)\,\phi\Big(\frac{x_2}{h}\Big)}.$$

- Compact support keeps each transfer local.
- The support width scales with h, so the interface is regularized over a few cells.
- Kernels trade support size, smoothness, translational invariance, and moment accuracy.

> δ_h is a grid-dependent approximation to the distribution $\delta^{(2)}$, not an ordinary pointwise approximation at the singularity.

WHAT PROPERTIES SHOULD ϕ POSSESS?

Core discrete moments

$$\sum_{j\in\mathbb{Z}} \phi(r-j) = 1, \qquad \sum_{j\in\mathbb{Z}} (r-j)\phi(r-j) = 0.$$

- zeroth moment: total-force conservation
- first moment: total-torque conservation and exact interpolation of linear fields

Desirable additional properties

- continuity and compact support
- even–odd balance for some kernels (Peskin's four-point kernel, built for a collocated grid with centered differences)
- constant sum of squares to improve approximate grid-translation invariance
- adequate regularity for the solver

Source: Peskin (2002), Secs. 5–6; Peskin's postulates are his eqs. (6.2)–(6.6).

THE EVEN–ODD CONDITION

Statement. For the one-dimensional kernel ϕ, the even- and odd-indexed grid points must each carry half the weight, for *every* marker position $r \in \mathbb{R}$:

$$\sum_{j \text{ even}} \phi(r-j) = \sum_{j \text{ odd}} \phi(r-j) = \tfrac{1}{2}$$

Why it is imposed (Peskin 2002, Sec. 6)

- It stops the even and odd grid points from decoupling in the spread and interpolate operators.
- Peskin's fluid solver uses the collocated centered difference D^0, whose stencil skips the centre point. The null space of the D^0 gradient contains every function that is constant on each of the 2^d even/odd "chains" of grid points (4 in 2D, 8 in 3D), not only the constants.
- Decoupled subgrids support checkerboard modes the interaction cannot see or damp. The even–odd condition gives every chain the same share of each marker's force and the same weight in its interpolated velocity, which suppresses point-to-point oscillations, especially under localized forces.
- The two sums add to $\sum_j \phi(r-j) = 1$: this is the split, stronger zeroth moment.

THE EVEN–ODD CONDITION (CONTINUED)

Not the sum-of-squares condition

$$\sum_{j\in\mathbb{Z}} \phi(r-j)^2 = C \text{ (const. in } r)$$

controls approximate translation invariance: coupling strength should not depend on where the marker sits in a cell. Neither condition implies the other: the cosine kernel satisfies both, the three-point kernel below satisfies only the sum of squares ($C = 1/2$), and a piecewise-linear three-point kernel can satisfy even–odd and first moment without a constant sum of squares.

Check at $r = 0$

- Peskin four-point: $\phi(0) = \frac{1}{2}$, $\phi(\pm 1) = \frac{1}{4}$, $\phi(\pm 2) = 0$. Even sum $\frac{1}{2} + 0 + 0 = \frac{1}{2}$; odd sum $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. ✓
- Roma–Peskin–Berger three-point: $\phi_3(0) = \frac{2}{3}$, $\phi_3(\pm 1) = \frac{1}{6}$. Even sum $\frac{2}{3} \neq \frac{1}{2}$: the condition fails.

Peskin's four-point kernel satisfies the even–odd condition. The three-point Roma–Peskin–Berger kernel is obtained by dropping it and keeping only $\sum_j \phi(r-j) = 1$, which narrows the support from four to three meshwidths (27 instead of 64 cells per marker in 3D). Peskin recommends it only with a fluid scheme that does not have the D^0 null-space problem, such as the staggered MAC scheme used by Roma, Peskin, and Berger (1999). On a staggered grid the even–odd condition is unnecessary; it is not true that a narrow stencil satisfies it automatically.

Sources: C. S. Peskin, *Acta Numerica* 11 (2002), Sec. 6, eqs. (6.4), (6.8) and the discussion of the three-point function; A. M. Roma, C. S. Peskin, M. J. Berger, *J. Comput. Phys.* 153 (1999) 509–534.

NUMERICAL ANALYSIS: THREE DIFFERENT QUESTIONS

Consistency	Stability	Convergence
Does the discrete equation approach the continuum equation as the mesh is refined?	Do perturbations remain controlled over the interval of interest?	Does the computed solution approach the exact solution?

> For a well-posed linear initial-value problem, consistency plus stability implies convergence. Nonlinear multiphysics calculations still require problem-specific analysis and verification.

NUMERICAL STABILITY: A SIMPLE TEST PROBLEM

Consider

$$y'(t) = -15y(t), \qquad y(0) = 1, \qquad y(t) = e^{-15t}.$$

Forward Euler gives

$$y^{n+1} = (1 - 15\Delta t)y^n.$$

Thus $g = 1 - 15\Delta t$, and absolute stability requires

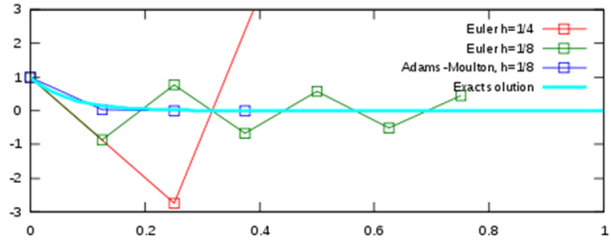
$$\boxed{|g| \leq 1 \quad \Longleftrightarrow \quad 0 \leq \Delta t \leq \frac{2}{15}}.$$

- $0 < \Delta t \leq 1/15$: monotone decay ($0 \leq g < 1$; at $\Delta t = 1/15$, $g = 0$ and the numerical solution is zero after one step).
- $1/15 < \Delta t < 2/15$: decaying sign oscillations ($\Delta t = 2/15$ is neutrally stable: $g = -1$, so the amplitude never decays).
- $\Delta t > 2/15$: growing instability.

STABILITY EXPERIMENT: WHAT THE PLOT SHOWS

- Forward Euler with $\Delta t = 1/4$ is unstable: $g = -2.75$.
- With $\Delta t = 1/8$, it is stable but oscillatory: $g = -0.875$.
- The trapezoidal rule is A-stable and gives a much better result here.

Try $\Delta t = 0.2,\ 0.1,\ 0.01$. Predict each behavior from $g = 1 - 15\Delta t$ before computing.



The plot solves $y' = -15y$, $y(0) = 1$; horizontal axis is t. The "Adams–Moulton, $h = 1/8$" curve is the trapezoidal rule (the two-stage Adams–Moulton method), with $g = (1 - 15h/2)/(1 + 15h/2) \approx 0.032$ at $h = 1/8$.

Figure: Wikimedia Commons, "StiffEquationNumericalSolvers.svg", by Berland (English Wikipedia), public domain.

WHY EXPLICIT IMMERSED-BOUNDARY UPDATES BECOME STIFF

Sources of stiffness

- large spring or target-point penalties
- bending operators with high spatial derivatives
- small Eulerian cells and tightly supported kernels
- rapidly varying active forcing

What controls the restriction

- the derivative of the structural force
- the coupled fluid response
- grid and marker spacing
- the chosen time integrator

> Stability is not inversely proportional to the instantaneous force magnitude. Linearized stiffness and the complete update operator determine whether perturbations grow.

ADVECTION, DIFFUSION, AND COUPLING IMPOSE DIFFERENT LIMITS

Representative multidimensional CFL

$$\Delta t \sum_{\alpha=1}^{2} \frac{\|u_\alpha\|_\infty}{\Delta x_\alpha} \le C_{\mathrm{CFL}}.$$

The admissible C_{CFL} depends on the advection discretization and integrator.

Explicit viscous restriction

$$\Delta t \lesssim C_\nu \frac{h^2}{\nu}.$$

Stiff structural forces add a third, method-dependent restriction that may be stronger than either fluid limit.

> A semi-implicit or implicit treatment can relax selected stiffness restrictions, but does not remove the need to verify accuracy.

CONVERGENCE AND VERIFICATION

For an error measured in a stated norm,

$$E(h) = \|u_h - u_{\mathrm{ref}}\|.$$

Refine separately

- Eulerian spacing h
- time step Δt
- marker spacing or interaction quadrature
- AMR hierarchy and domain size

Measure relevant outputs

- velocity, pressure, and deformation
- integrated force and moment
- leakage or enclosed-area change
- application-specific quantities

> A stable computation can converge to a poorly resolved discrete solution. Stability and accuracy answer different questions.

HEAT EQUATION: ONE LABORATORY FOR SPACE AND TIME ERROR

For $0 < x < 1$,

$$u_t = D\, u_{xx}.$$

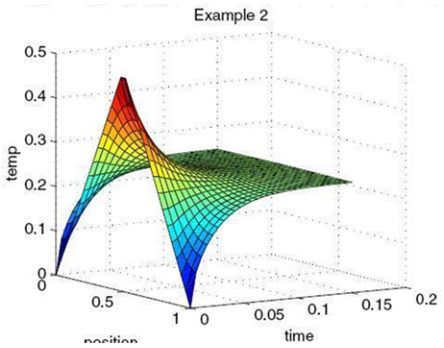
Take prescribed initial data and, for this laboratory, $u(0,t) = u(1,t) = 0$ and $D = 1$ in code units. Forward time, centered space gives

$$u_j^{n+1} = u_j^n + r\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right),$$

with mesh ratio $r = D\Delta t/\Delta x^2$. For this one-dimensional scheme,

$$\boxed{0 \le r \le \frac{1}{2}}$$

is the von Neumann stability condition.



Diffusion smooths spatial temperature variation

This illustration ("Example 2") has end values that rise from 0: it corresponds to insulated (zero-flux) ends, so the triangular profile relaxes to its mean ≈ 0.22; the laboratory itself uses $u = 0$ at both ends, so its solution decays to zero.

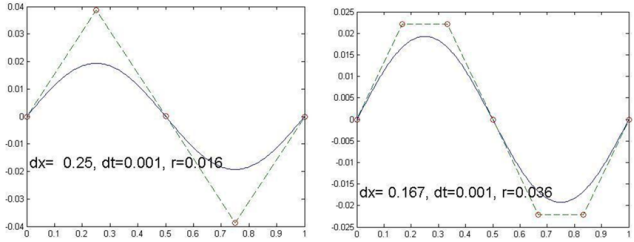
Instructor's course materials (MATLAB heat-equation lab output).

HEAT EQUATION: SPATIAL-REFINEMENT EXPERIMENT

Using your own script, with $D = 1$ in code units:

1. hold the final time at 0.1;
2. set $T = 100$ time steps;
3. try $n = 3, 5, 10, 20$;
4. compare in one stated error norm.

Check $r = D\Delta t/\Delta x^2$ for every run. A refinement study is interpretable only while time error is stable and sufficiently small.



Here n counts interior points, so $\Delta x = 1/(n+1)$ and $\Delta t = 0.1/100 = 10^{-3}$: $r = 10^{-3}(n+1)^2$. The panels show $n = 3$ ($\Delta x = 0.25$, $r = 0.016$) and $n = 5$ ($\Delta x \approx 0.167$, $r = 0.036$); even $n = 20$ gives $r = 0.441 \le \frac{1}{2}$, so every run in this study is stable. Solid: exact solution $e^{-4\pi^2 t}\sin(2\pi x)$ at $t = 0.1$; circles: FTCS.

Instructor's course materials (MATLAB heat-equation lab output).

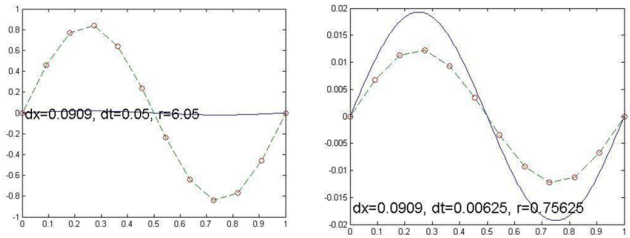
HEAT EQUATION: TIME-STEP STABILITY AND CONVERGENCE

Fix $n = 10$ and run to time 0.1:

$$T \in \{2, 4, 8, 16, 32, 64\},$$

$$\Delta t = \frac{0.1}{T}.$$

- Predict stable cases using $r \leq 1/2$.
- Among stable cases, measure temporal error.
- Do not estimate an order from unstable runs.



Large r can create amplification rather than diffusion

Panels: $T = 2$ ($r = 6.05$) and $T = 16$ ($r \approx 0.76$); read them with the next slide.

Instructor's course materials (MATLAB heat-equation lab output).

HEAT EQUATION: READING THE TIME-STEP PLOTS

With $n = 10$ interior points, $\Delta x = 1/11$ and $\Delta t = 0.1/T$, so $r = 12.1/T$. FTCS multiplies the discrete mode $\sin(k\pi x_j)$ by

$$g_k = 1 - 4r \sin^2\Big(\frac{k\pi\Delta x}{2}\Big) \quad \text{per step,} \qquad k = 1, \ldots, n.$$

Left panel: $T = 2$**,** $r = 6.05$

- The initial sine $\sin(2\pi x)$ is the single mode $k = 2$: $g_2 \approx -0.92$.
- Two steps leave amplitude $0.92^2 \approx 0.85$; the exact amplitude is $e^{-4\pi^2(0.1)} \approx 0.019$.
- Higher modes have $|g_k|$ up to ≈ 23 and would amplify any perturbation.

Right panel: $T = 16$**,** $r \approx 0.76$

- $r > \frac{1}{2}$: formally unstable.
- Only the stable smooth mode is excited ($g_2 \approx 0.76$, giving amplitude ≈ 0.012), so the visible error is excess damping.
- Round-off in the highest mode ($g_{10} \approx -1.96$) grows, but from $\sim 10^{-16}$ it stays invisible over 16 steps.

> Stability is about *all* modes. A smooth initial condition can hide an instability for a while; a longer run, a rougher initial condition, or round-off will expose it. Only $T = 32$ ($r \approx 0.38$) and $T = 64$ ($r \approx 0.19$) in this list satisfy $r \leq \frac{1}{2}$.

RATE OF CONVERGENCE OF AN ITERATIVE SEQUENCE

Suppose $x_k \to L$. If

$$\lim_{k\to\infty} \frac{|x_{k+1} - L|}{|x_k - L|^q} = C, \qquad C > 0,$$

then q is the *order of convergence* of the sequence.

- $q = 1$, with $0 < C < 1$: linear convergence.
- $q = 2$: quadratic convergence.
- $q = 3$: cubic convergence.

> This definition concerns iterations k. Mesh-convergence order uses a different limiting parameter: $h \to 0$.

OBSERVED ORDER OF A SPATIAL OR TEMPORAL DISCRETIZATION

If

$$E(h) = Ch^p + o(h^p),$$

then p is the asymptotic discretization order. With refinement ratio two,

$$\boxed{p_{\mathrm{obs}} \approx \frac{\log(E(h)/E(h/2))}{\log 2}}.$$

- Use the same norm, final time, geometry, and physical parameters.
- Keep other error sources smaller than the one being measured.
- Verify that p_{obs} stabilizes across multiple refinement pairs.

RESOLUTION AND INTERFACE ACCURACY

Interaction resolution

- For nodal IB, current marker spacing of roughly $h/2$ is a common empirical watertightness choice, not a theorem.
- Large deformation can open gaps, so monitor spacing over time.
- In IBFE, adequately resolved interaction quadrature helps prevent leakage and need not coincide with structural nodes.

Expected accuracy

- Singular interface forces generally yield first-order global behavior for classical regularized-delta IB schemes.
- Pressure and velocity derivatives may jump across the interface.
- Smoother special cases can show nearly second-order convergence in selected fields and norms.

> There is no universal mesh ratio, grid size, or time step. Report the variable, norm, and refinement path.

WHAT IBAMR PROVIDES

IBAMR is an open-source C++ research framework

- parallel solvers for incompressible flow
- several immersed-boundary and immersed finite-element formulations
- nested, block-structured Cartesian AMR
- data transfer and synchronization across refinement levels

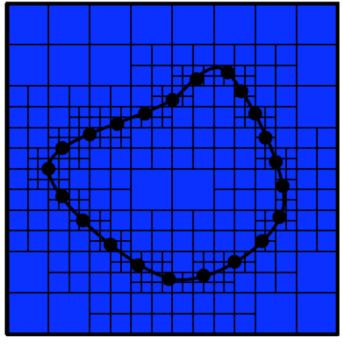
> AMR accelerates a calculation only when hierarchy overhead is outweighed by avoiding a uniformly fine grid. It does not replace stability and convergence studies.

ibamr.github.io

IBAMR: NONCONFORMING STRUCTURE ON AN ADAPTIVE GRID

- Eulerian variables live on locally refined Cartesian patches.
- The Lagrangian structure moves independently of patch boundaries.
- In the standard configuration, interaction couples the structure to the finest Eulerian level that covers it.
- Tagging and regridding must keep fine coverage around the moving structure.

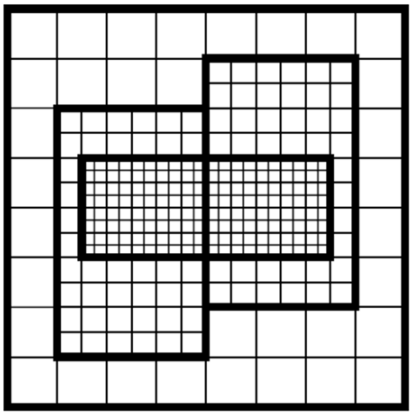
> The Lagrangian mesh does not acquire the Eulerian hierarchy's coarse–fine levels.



Fine Eulerian cells track an immersed curve

Adaptive refinement around an immersed boundary, after IBAMR documentation (B. E. Griffith); compare Griffith et al., *J. Comput. Phys.* 223 (2007). Source to be confirmed.

ADAPTIVE MESH REFINEMENT IN PRACTICE



Nested Cartesian patches at several resolutions

Block-structured AMR schematic after M. J. Berger and P. Colella, *J. Comput. Phys.* 82 (1989) 64–84. Source to be confirmed.

- Tag cells using problem-relevant indicators: structure location, vorticity, or solution gradients.
- Synchronize fluxes and data across coarse–fine interfaces.
- Regrid often enough to follow moving features.
- Quote the finest spacing, refinement ratios, tagging criteria, and regrid interval.
- Verify that changing the hierarchy does not change the quantity of interest.

Mathematical details of the coupling

$$\boldsymbol{F} \longrightarrow \boldsymbol{f} \longrightarrow \boldsymbol{u} \longrightarrow \boldsymbol{U} = \partial_t \boldsymbol{X}.$$

MATHEMATICAL FORMULATION: NOTATION AND ASSUMPTIONS

- $\boldsymbol{x} = (x_1, x_2)$: fixed spatial coordinate
- q: material coordinate on $\mathcal{U}$
- $\boldsymbol{X}(q,t)$: position of material point q
- $\boldsymbol{u}(\boldsymbol{x},t)$: Eulerian velocity
- $\boldsymbol{F}(q,t)$: Lagrangian force per unit q
- $\boldsymbol{f}(\boldsymbol{x},t)$: Eulerian force density

Scope

- incompressible Newtonian fluid
- constant ρ, μ
- massless elastic curve
- no slip
- force from $E[\boldsymbol{X},t]$

> A volumetric IBFE structure requires its own material constitutive law; structural incompressibility is not assumed here.

LAGRANGIAN–EULERIAN INTERACTION CONSERVES FORCE AND TORQUE

The statement has content only for the *regularized* kernel: spread with $\boldsymbol{f_i} = \sum_m \boldsymbol{F}_m \, \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) \, w_m$. Then

$$\sum_{\boldsymbol{i}} \boldsymbol{f_i} \, h^2 = \sum_m \boldsymbol{F}_m \, w_m \qquad \underbrace{\left[\sum_{\boldsymbol{i}} \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) \, h^2 \right]}_{= \prod_{\alpha=1}^{2} \sum_{j \in \mathbb{Z}} \phi(r_\alpha^m - j) \; = \; 1} \qquad = \sum_m \boldsymbol{F}_m \, w_m,$$

where $r_\alpha^m = (X_{m,\alpha})/h$. The bracket collapses to 1 because ϕ satisfies the zeroth moment $\sum_j \phi(r - j) = 1$ for all r; nothing else forces it.

Here the grid nodes are $\boldsymbol{x_i} = \boldsymbol{i}h$ and ϕ is even, so $h^2 \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) = \phi(r_1^m - i_1) \, \phi(r_2^m - i_2)$. On a staggered or cell-centred grid, shift r by the half-cell offset of that component's nodes; the argument is unchanged. The sums are over the whole (unbounded or periodic) grid.

INTERACTION CONSERVES FORCE AND TORQUE (CONTINUED: TORQUE)

With the first moment $\sum_j (r-j)\phi(r-j) = 0$, the same computation gives $\sum_{\boldsymbol{i}} \boldsymbol{x_i}\, \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) h^2 = \boldsymbol{X}_m$, so the torques also agree:

$$\sum_{\boldsymbol{i}} \boldsymbol{x_i} \times \boldsymbol{f_i}\, h^2 = \sum_m \boldsymbol{X}_m \times \boldsymbol{F}_m\, w_m.$$

The missing step. Subtracting the first moment from r times the zeroth moment gives

$$\sum_j j\, \phi(r-j) = r \sum_j \phi(r-j) - \sum_j (r-j)\phi(r-j) = r,$$

so in each coordinate $\sum_{i_\alpha} (i_\alpha h)\, \phi(r^m_\alpha - i_\alpha) = h\, r^m_\alpha = X_{m,\alpha}$, while the other factor sums to 1. Then $\sum_{\boldsymbol{i}} \boldsymbol{x_i} \times \boldsymbol{f_i} h^2 = \sum_m \left[\sum_{\boldsymbol{i}} \boldsymbol{x_i} \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) h^2\right] \times \boldsymbol{F}_m w_m$. In 2D the torque is the scalar $x_1 f_2 - x_2 f_1$. (Peskin 2002, eqs. (5.5), (5.8), (6.7).)

> For the exact Dirac delta both identities are automatic; the moment conditions are what recover them at finite h.

SMOOTHED APPROXIMATIONS TO THE DIRAC DELTA FUNCTION

For $\delta_h(\boldsymbol{x}) = h^{-2}\phi(x_1/h)\phi(x_2/h)$, a useful checklist is:

1. **compact support** for local interaction;
2. **zeroth moment** $\sum_j \phi(r-j) = 1$;
3. **first moment** $\sum_j (r-j)\phi(r-j) = 0$;
4. **regularity** compatible with the solver;
5. **translation-control conditions**, such as a constant sum of squares, when desired.

> No compactly supported grid kernel is exactly translation invariant for arbitrary marker motion. Kernel choice influences smoothness, accuracy, and cost.

CLASSICAL IMMERSED-BOUNDARY CHECKLIST

1. State the interface dimension, material measure, and structural assumptions.
2. Define $\boldsymbol{F} = -\delta E / \delta \boldsymbol{X}$, including target penalties if used.
3. Include Lagrangian quadrature weights in discrete spreading.
4. Use one kernel consistently so spreading and interpolation are adjoint.
5. State time levels and check advection, diffusion, and coupling stiffness.
6. Refine space, time, and interaction resolution independently.

> Force, velocity, and power must agree in the Eulerian and Lagrangian descriptions.

QUESTIONS?

$$\boldsymbol{F} \rightarrow \boldsymbol{f} \rightarrow \boldsymbol{u} \rightarrow \boldsymbol{U} = \partial_t \boldsymbol{X}$$

PRIMARY REFERENCES AND OFFICIAL RESOURCES

Primary references

- C. S. Peskin, *The immersed boundary method*, *Acta Numerica* 11 (2002), 479–517. doi:10.1017/S0962492902000077
- B. E. Griffith and X. Luo, *Hybrid finite difference/finite element immersed boundary method*, *Int. J. Numer. Methods Biomed. Eng.* 33 (2017), e2888. doi:10.1002/cnm.2888
- B. E. Griffith and N. A. Patankar, *Immersed methods for fluid–structure interaction*, *Annu. Rev. Fluid Mech.* 52 (2020), 421–448. doi:10.1146/annurev-fluid-010719-060228

Official resources

- IBAMR documentation
- IBAMR source

> The repository's seminal papers support the formulation and numerical claims.

Adapted from: the continuum formulation, spreading/interpolation, moment conditions, and conservation identities follow Peskin (2002), especially Secs. 5–6; the interaction-operator and IBFE remarks follow Griffith and Luo (2017) and Griffith and Patankar (2020). Further sources on the next slide.

REFERENCES (CONTINUED)

Method and kernels

- C. S. Peskin, *Numerical analysis of blood flow in the heart*, *J. Comput. Phys.* 25 (1977), 220–252. doi:10.1016/0021-9991(77)90100-0
- A. M. Roma, C. S. Peskin, and M. J. Berger, *An adaptive version of the immersed boundary method*, *J. Comput. Phys.* 153 (1999), 509–534. doi:10.1006/jcph.1999.6293
- F. H. Harlow and J. E. Welch, *Numerical calculation of time-dependent viscous incompressible flow of fluid with free surface*, *Phys. Fluids* 8 (1965), 2182–2189 (MAC grid). doi:10.1063/1.1761178
- R. J. LeVeque, *Finite Difference Methods for Ordinary and Partial Differential Equations*, SIAM (2007): stiffness, absolute stability, von Neumann analysis, consistency + stability $\Rightarrow$ convergence. doi:10.1137/1.9780898717839

Scales

- S. Vogel, *Life in Moving Fluids*, 2nd ed., Princeton University Press (1994): Reynolds-number orders of magnitude.

Adaptive refinement

- M. J. Berger and P. Colella, *Local adaptive mesh refinement for shock hydrodynamics*, *J. Comput. Phys.* 82 (1989), 64–84. doi:10.1016/0021-9991(89)90035-1
- B. E. Griffith, R. D. Hornung, D. M. McQueen, and C. S. Peskin, *An adaptive, formally second order accurate version of the immersed boundary method*, *J. Comput. Phys.* 223 (2007), 10–49.

Figures

- Stability plot: Wikimedia Commons, “StiffEquationNumericalSolvers.svg” (Berland), public domain.
- Heat-equation plots: instructor’s MATLAB lab output. Spreading/interpolation schematics: instructor’s course notes. IBAMR and AMR schematics: after Griffith (IBAMR) and Berger–Colella; sources to be confirmed.